Graph Paths II

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a directed weighted graph having n nodes and m edges. Your task is to calculate the minimum path length from node 1 to node n with exactly k edges.

Input

The first input line contains three integers n, m and k: the number of nodes and edges, and the length of the path. The nodes are numbered $1, 2, \dots, n$.

Then, there are m lines describing the edges. Each line contains three integers a, b and c: there is an edge from node a to node b with weight c.

Output

Print the minimum path length. If there are no such paths, print -1.

Constraints

- $1 \le n \le 100$
- $1 \le m \le n(n-1)$
- $1 \le k \le 10^9$
- $1 \le a, b \le n$
- $1 \le c \le 10^9$

Example

Input	Output
3 4 8 1 2 5 2 3 4 3 1 1 3 2 2	27